



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
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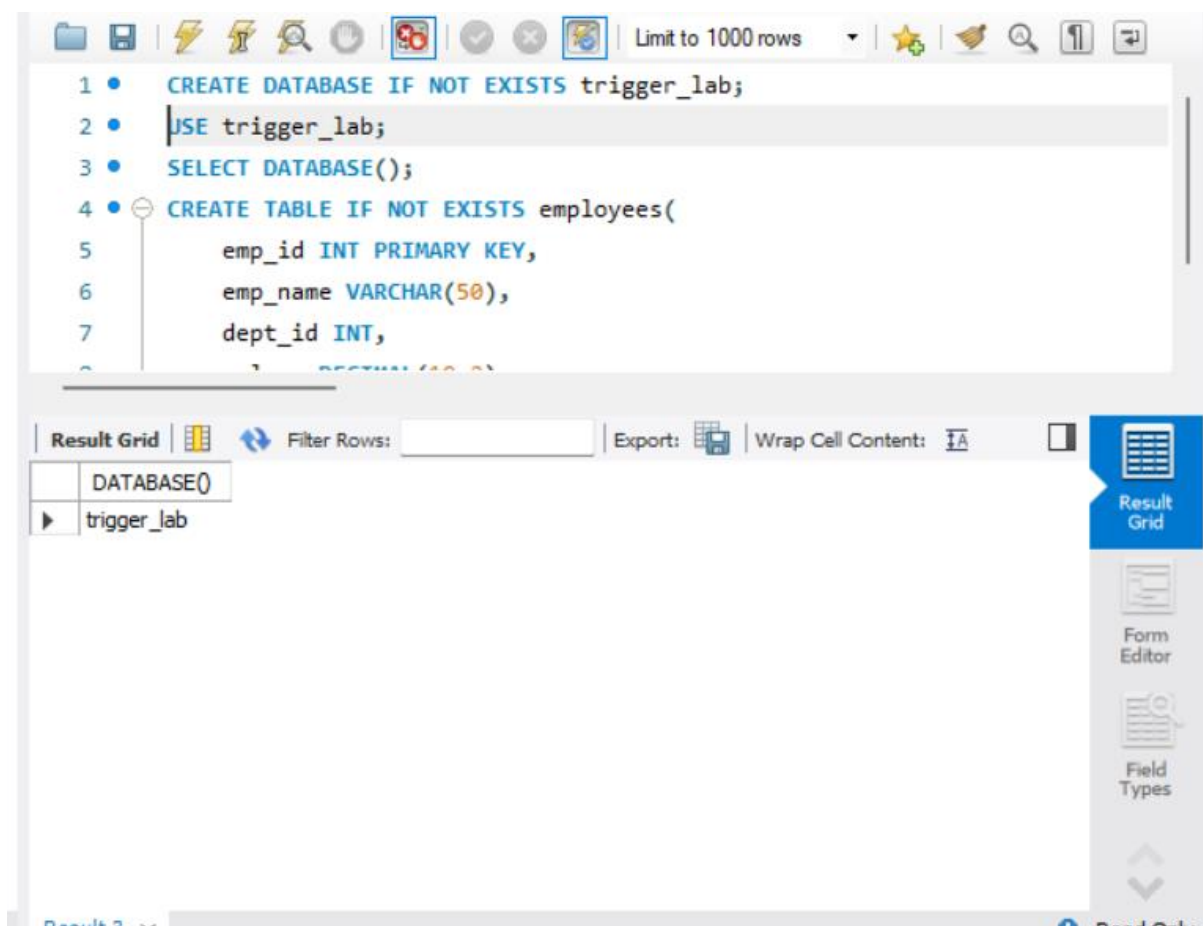
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MYSQL TRIGGERS

Practical Examples and Student Exercises using MySQL Workbench



```

4 • CREATE TABLE IF NOT EXISTS employees(
5     emp_id INT PRIMARY KEY,
6     emp_name VARCHAR(50),
7     dept_id INT,
8     salary DECIMAL(10,2)
9 );
10

```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



Result Grid

Form Editor

DATABASE()
trigger_lab

Limit to 1000 rows

```

10 • INSERT INTO employees VALUES
11     (101, 'Rahul', 1, 50000),
12     (102, 'Priya', 2, 45000),
13     (103, 'Kiran', 1, 55000),
14     (104, 'Anu', 2, 48000),
15     (105, 'Vijay', 1, 60000);
16 • SELECT * FROM employees;

```

Result Grid



Filter Rows:

Edit:



Export/Import:



Result Grid

Form Editor

Field Types

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	60000.00
	102	Priya	2	45000.00
	103	Kiran	1	55000.00
	104	Anu	2	48000.00
	105	Vijay	1	60000.00
	108	Meena	1	30000.00
•	NULL	NULL	NULL	NULL

Result 7

employees 8 x

Apply

```

17 • CREATE TABLE IF NOT EXISTS employee_audit(
18     audit_id INT AUTO_INCREMENT PRIMARY KEY,
19     emp_id INT,
20     emp_name VARCHAR(50),
21     action_type VARCHAR(20),
22     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
23 );

```

Result Grid | Filter Rows: | Edit: | Export/Import:

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	60000.00
	102	Priya	2	45000.00
	103	Kiran	1	55000.00
	104	Anu	2	48000.00
	105	Vijay	1	60000.00
	108	Meena	1	30000.00
*	NULL	NULL	NULL	NULL

Limit to 1000 rows

```

17 • CREATE TABLE IF NOT EXISTS employee_audit(
18     audit_id INT AUTO_INCREMENT PRIMARY KEY,
19     emp_id INT,
20     emp_name VARCHAR(50),
21     action_type VARCHAR(20),
22     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
23 );

```

Result Grid | Filter Rows: | Edit: | Export/Import:

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-11 21:22:41
	2	106	Arjun	DELETE	2026-08-11 21:23:56
	3	106	Arjun	INSERT	2026-08-11 21:24:31
	4	106	Arjun	DELETE	2026-08-11 21:24:31
	5	106	Arjun	INSERT	2026-08-11 21:25:17
	6	106	Arjun	DELETE	2026-08-11 21:25:17
	7	106	Arjun	INSERT	2026-08-11 21:25:45
	8	106	Arjun	DELETE	2026-08-11 21:25:45
	9	101	Rahul	UPDATE	2026-08-11 21:25:45
	10	106	Arjun	INSERT	2026-08-11 21:26:09
	11	106	Arjun	DELETE	2026-08-11 21:26:09
	12	101	Rahul	UPDATE	2026-08-11 21:26:09

Result Grid
Form Editor
Field Types

```

24 • SELECT * FROM employee_audit;
25 • DROP TRIGGER IF EXISTS after_employee_insert;

```

Result Grid					
Filter Rows:					
Edit:   					
Export/Import:  					
audit_id	emp_id	emp_name	action_type	action_time	
1	106	Arjun	INSERT	2026-08-11 21:22:41	
2	106	Arjun	DELETE	2026-08-11 21:23:56	
3	106	Arjun	INSERT	2026-08-11 21:24:31	
4	106	Arjun	DELETE	2026-08-11 21:24:31	
5	106	Arjun	INSERT	2026-08-11 21:25:17	
6	106	Arjun	DELETE	2026-08-11 21:25:17	
7	106	Arjun	INSERT	2026-08-11 21:25:45	
8	106	Arjun	DELETE	2026-08-11 21:25:45	
9	101	Rahul	UPDATE	2026-08-11 21:25:45	
10	106	Arjun	INSERT	2026-08-11 21:26:09	
11	106	Arjun	DELETE	2026-08-11 21:26:09	
12	101	Rahul	UPDATE	2026-08-11 21:26:09	

Result Grid					
Filter Rows:					
Edit:   					
Export/Import:  					
audit_id	emp_id	emp_name	action_type	action_time	
1	106	Arjun	INSERT	2026-08-11 21:22:41	
2	106	Arjun	DELETE	2026-08-11 21:23:56	
3	106	Arjun	INSERT	2026-08-11 21:24:31	
4	106	Arjun	DELETE	2026-08-11 21:24:31	
5	106	Arjun	INSERT	2026-08-11 21:25:17	
6	106	Arjun	DELETE	2026-08-11 21:25:17	
7	106	Arjun	INSERT	2026-08-11 21:25:45	
8	106	Arjun	DELETE	2026-08-11 21:25:45	
9	101	Rahul	UPDATE	2026-08-11 21:25:45	
10	106	Arjun	INSERT	2026-08-11 21:26:09	
11	106	Arjun	DELETE	2026-08-11 21:26:09	
12	101	Rahul	UPDATE	2026-08-11 21:26:09	

```

33      (emp_id, emp_name, action_type)
34      VALUES
35      (NEW.emp_id, NEW.emp_name, 'INSERT');
36  END $$
37
38  DELIMITER ;
39 • INSERT INTO employees VALUES (106, 'Arjun', 1, 52000);

```

```

34     VALUES
35     (NEW.emp_id, NEW.emp_name, 'INSERT');
36 END $$
37
38 DELIMITER ;
39 • INSERT INTO employees VALUES (106, 'Arjun', 1, 52000);
40 • DROP TRIGGER IF EXISTS after_employee_delete;

```

Result Grid					
	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-11 21:22:41
	2	106	Arjun	DELETE	2026-08-11 21:23:56
	3	106	Arjun	INSERT	2026-08-11 21:24:31
	4	106	Arjun	DELETE	2026-08-11 21:24:31
	5	106	Arjun	INSERT	2026-08-11 21:25:17
	6	106	Arjun	DELETE	2026-08-11 21:25:17
	7	106	Arjun	INSERT	2026-08-11 21:25:45
	8	106	Arjun	DELETE	2026-08-11 21:25:45
	9	101	Rahul	UPDATE	2026-08-11 21:25:45
	10	106	Arjun	INSERT	2026-08-11 21:26:09
	11	106	Arjun	DELETE	2026-08-11 21:26:09
	12	101	Rahul	UPDATE	2026-08-11 21:26:09

```

46 BEGIN
47     INSERT INTO employee_audit
48     (emp_id, emp_name, action_type)
49     VALUES
50     (OLD.emp_id, OLD.emp_name, 'DELETE');
51 END $$
52
53 DELIMITER ;

```

Result Grid					
	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-11 21:22:41
	2	106	Arjun	DELETE	2026-08-11 21:23:56
	3	106	Arjun	INSERT	2026-08-11 21:24:31
	4	106	Arjun	DELETE	2026-08-11 21:24:31
	5	106	Arjun	INSERT	2026-08-11 21:25:17
	6	106	Arjun	DELETE	2026-08-11 21:25:17
	7	106	Arjun	INSERT	2026-08-11 21:25:45
	8	106	Arjun	DELETE	2026-08-11 21:25:45
	9	101	Rahul	UPDATE	2026-08-11 21:25:45
	10	106	Arjun	INSERT	2026-08-11 21:26:09
	11	106	Arjun	DELETE	2026-08-11 21:26:09
	12	101	Rahul	UPDATE	2026-08-11 21:26:09


```

54 • DELETE FROM employees
55 WHERE emp_id = 106;

```

Result Grid					
Filter Rows:					
Edit:					
Export/Import					
	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-11 21:22:41
	2	106	Arjun	DELETE	2026-08-11 21:23:56
	3	106	Arjun	INSERT	2026-08-11 21:24:31
	4	106	Arjun	DELETE	2026-08-11 21:24:31
	5	106	Arjun	INSERT	2026-08-11 21:25:17
	6	106	Arjun	DELETE	2026-08-11 21:25:17
	7	106	Arjun	INSERT	2026-08-11 21:25:45
	8	106	Arjun	DELETE	2026-08-11 21:25:45
	9	101	Rahul	UPDATE	2026-08-11 21:25:45
	10	106	Arjun	INSERT	2026-08-11 21:26:09
	11	106	Arjun	DELETE	2026-08-11 21:26:09
	12	101	Rahul	UPDATE	2026-08-11 21:26:09

```

56 • SELECT * FROM employee_audit;

```

Result Grid					
Filter Rows:					
Edit:					
Export/Import					
	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-11 21:22:41
	2	106	Arjun	DELETE	2026-08-11 21:23:56
	3	106	Arjun	INSERT	2026-08-11 21:24:31
	4	106	Arjun	DELETE	2026-08-11 21:24:31
	5	106	Arjun	INSERT	2026-08-11 21:25:17
	6	106	Arjun	DELETE	2026-08-11 21:25:17
	7	106	Arjun	INSERT	2026-08-11 21:25:45
	8	106	Arjun	DELETE	2026-08-11 21:25:45
	9	101	Rahul	UPDATE	2026-08-11 21:25:45
	10	106	Arjun	INSERT	2026-08-11 21:26:09
	11	106	Arjun	DELETE	2026-08-11 21:26:09
	12	101	Rahul	UPDATE	2026-08-11 21:26:09

Result 41 employees 42 employee_audit 43 employee_audit 44 ▼

```

60 • CREATE TRIGGER after_employee_update
61 AFTER UPDATE ON employees
62 FOR EACH ROW
63 BEGIN
64     INSERT INTO employee_audit
65     (emp_id, emp_name, action_type)
66     VALUES

```

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-11 21:22:41
	2	106	Arjun	DELETE	2026-08-11 21:23:56
	3	106	Arjun	INSERT	2026-08-11 21:24:31
	4	106	Arjun	DELETE	2026-08-11 21:24:31
	5	106	Arjun	INSERT	2026-08-11 21:25:17
	6	106	Arjun	DELETE	2026-08-11 21:25:17
	7	106	Arjun	INSERT	2026-08-11 21:25:45
	8	106	Arjun	DELETE	2026-08-11 21:25:45
	9	101	Rahul	UPDATE	2026-08-11 21:25:45
	10	106	Arjun	INSERT	2026-08-11 21:26:09
	11	106	Arjun	DELETE	2026-08-11 21:26:09
	12	101	Rahul	UPDATE	2026-08-11 21:26:09

Result 56 employees 57 employee_audit 58 employee_audit 59 × Apply

```

82      IF NEW.salary < 20000 THEN
83          SIGNAL SQLSTATE '45000'
84          SET MESSAGE_TEXT = 'Salary cannot be less than 20000';
85      END IF;
86  END $$
87
88  DELIMITER ;

```

Result Grid					
	audit_id	emp_id	emp_name	action_type	action_time
1	1	106	Arjun	INSERT	2026-08-11 21:22:41
2	2	106	Arjun	DELETE	2026-08-11 21:23:56
3	3	106	Arjun	INSERT	2026-08-11 21:24:31
4	4	106	Arjun	DELETE	2026-08-11 21:24:31
5	5	106	Arjun	INSERT	2026-08-11 21:25:17
6	6	106	Arjun	DELETE	2026-08-11 21:25:17
7	7	106	Arjun	INSERT	2026-08-11 21:25:45
8	8	106	Arjun	DELETE	2026-08-11 21:25:45
9	9	101	Rahul	UPDATE	2026-08-11 21:25:45
10	10	106	Arjun	INSERT	2026-08-11 21:26:09
11	11	106	Arjun	DELETE	2026-08-11 21:26:09
12	12	101	Rahul	UPDATE	2026-08-11 21:26:09

Result 78 employees 79 employee_audit 80 employee_audit 81 employee_audit 82 Apply

Output :


```

89 • INSERT INTO employees
90 VALUES (107, 'Ravi', 1, 15000);

```

Result Grid					
Filter Rows:					
Edit:					
Export/Import:					
	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-11 21:22:41
	2	106	Arjun	DELETE	2026-08-11 21:23:56
	3	106	Arjun	INSERT	2026-08-11 21:24:31
	4	106	Arjun	DELETE	2026-08-11 21:24:31
	5	106	Arjun	INSERT	2026-08-11 21:25:17
	6	106	Arjun	DELETE	2026-08-11 21:25:17
	7	106	Arjun	INSERT	2026-08-11 21:25:45
	8	106	Arjun	DELETE	2026-08-11 21:25:45
	9	101	Rahul	UPDATE	2026-08-11 21:25:45
	10	106	Arjun	INSERT	2026-08-11 21:26:09
	11	106	Arjun	DELETE	2026-08-11 21:26:09
	12	101	Rahul	UPDATE	2026-08-11 21:26:09

Result 84 employees 85 employee_audit 86 employee_audit 87 [employee_audit](#)

```

89 • INSERT INTO employees
90 VALUES (107, 'Ravi', 1, 15000);
91 • SELECT * FROM employees;

```

Result Grid				
Filter Rows:				
Edit:				
	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	60000.00
	102	Priya	2	45000.00
	103	Kiran	1	55000.00
	104	Anu	2	48000.00
	105	Vijay	1	60000.00
	108	Meena	1	30000.00
•	NULL	NULL	NULL	NULL

```

90     VALUES (107, 'Ravi', 1, 15000);
91 •   SELECT * FROM employees;
92 •   INSERT INTO employees VALUES (108, 'Meena', 1, 30000);

```

Result Grid | Filter Rows: | Edit: | Export/Import:

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	60000.00
	102	Priya	2	45000.00
	103	Kiran	1	55000.00
	104	Anu	2	48000.00
	105	Vijay	1	60000.00
	108	Meena	1	30000.00
*	NULL	NULL	NULL	NULL

```

101     IF NEW.salary < OLD.salary THEN
102         SIGNAL SQLSTATE '45000'
103         SET MESSAGE_TEXT = 'Salary reduction is not allowed';
104     END IF;
105 END $$
106
107 DELIMITER ;

```

Result Grid | Filter Rows: | Edit: | Export/Import:

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	60000.00
	102	Priya	2	45000.00
	103	Kiran	1	55000.00
	104	Anu	2	48000.00
	105	Vijay	1	60000.00
	108	Meena	1	30000.00
*	NULL	NULL	NULL	NULL

Result Grid
Form Editor
Field Types

```

107 DELIMITER ;
108 • UPDATE employees
109 SET salary = 40000
110 WHERE emp_id = 101;
111 • SELECT * FROM employees;
112 • SELECT * FROM employee_audit;
113 • SHOW TRIGGERS;
...

```

Result Grid Filter Rows: Export: Wrap Cell Content:

Trigger	Event	Table	Statement
check_salary	INSERT	employees	BEGIN IF NEW.salary < 20000 THEN SI...
after_employee_insert	INSERT	employees	BEGIN INSERT INTO employee_audit (emp...
prevent_salary_reduction	UPDATE	employees	BEGIN IF NEW.salary < OLD.salary THEN ...
after_employee_update	UPDATE	employees	BEGIN INSERT INTO employee_audit (emp...
after_employee_delete	DELETE	employees	BEGIN INSERT INTO employee_audit (emp...

Result Grid

Form Editor

Field Types

employees 160 employees 161 employees 162 employee_audit 163 Result 164 Read Only